# Qian 2020 breast cancer — analysis summary RUN 3 OF 3

## GO / NO-GO GATES

### Gate 1 — mixing & continuity → NO-GO

Harmony on sample cannot integrate the compartment. Across θ 2–20 (confirmatory to 50) × n_pcs 20–40 the regenerated embedding keeps the dominant patient (sample 54, 65% of malignant cells) as a segregated territory: at low θ a literal disconnected component (2,675 vs 2,143 cells), at high θ a lobe the kNN graph merges only as a connectivity artifact. Per-neighbourhood iLISI stays ≤1.5 of 2.26; sample-54 cells keep 92% same-sample 30-NN (chance 65%). Dropping sample 54, the other 6 integrate well (iLISI 3.4/4.86) — the obstruction is the dominant patient, not a generic batch.

### Gate 2 — module geography → NA

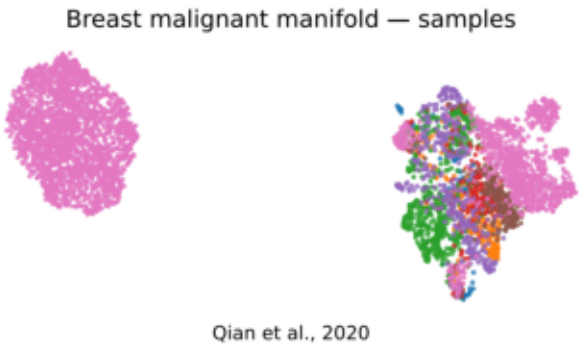
Not reached. Gate 1 NO-GO stops the chain before a manifold/signatures are fit, so module geography is not evaluated (manifold / signatures / correlation / modules notebooks are not produced, per the deliverable spec).

### Gate 3 — cross-repression → NA

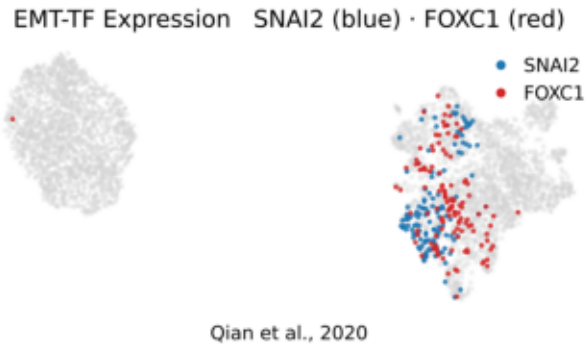
Not reached — no trajectory, hence no bifurcation and no hub cross-repression test.

## STEP FIGURES

**Malignant manifold — samples**




**EMT-TF expression**




---

**Dataset.**
Breast cancer (Qian 2020), 10x, 4,818 malignant cells over 7 patients; sample 54 alone is 65% of the malignant compartment (one dominant patient). No sample dropped (all 7 pass the 10-cell floor).

**Verdict.**
The embryonic pattern does NOT reproduce here — the malignant compartment cannot be integrated into a continuous manifold: a single dominant patient forms an irreconcilable segregated territory at every integration strength, so no cell-state trajectory is fit. Gate 1 NO-GO / Gate 2 NA / Gate 3 NA.

## Parameter choices

| | | | |
|---|---|---|---|
| **Sample threshold** | ≥10 malignant cells/sample; all 7 pass → 4,818 cells | **Overdispersed genes** | 1,225 (scFates find_overdispersed); no sc.pp.scale |
| **Integration** | Harmony on sample; θ 2/5/10/20 (+50); best mixing θ=5 | **Embedding** | n_pcs swept 20/30/40 (committed 30); n_neighbors=30; spread=1.3 |
| **Gate-1 metric** | per-neighbourhood iLISI (max 2.26) + UMAP-radius-graph components | **Module means (raw)** | Gast −0.020 · pSom +0.031 · Crest +0.013 |
| **Geometry** | NA — no trajectory fit (Gate 1 NO-GO) | **EMT-TFs shown** | SNAI2 (11.3%) · FOXC1 (10.4%); FOXC/PRRX in panel |